



EECS 570 Programming Assignment 1 (Continued)

September 5th, 2025

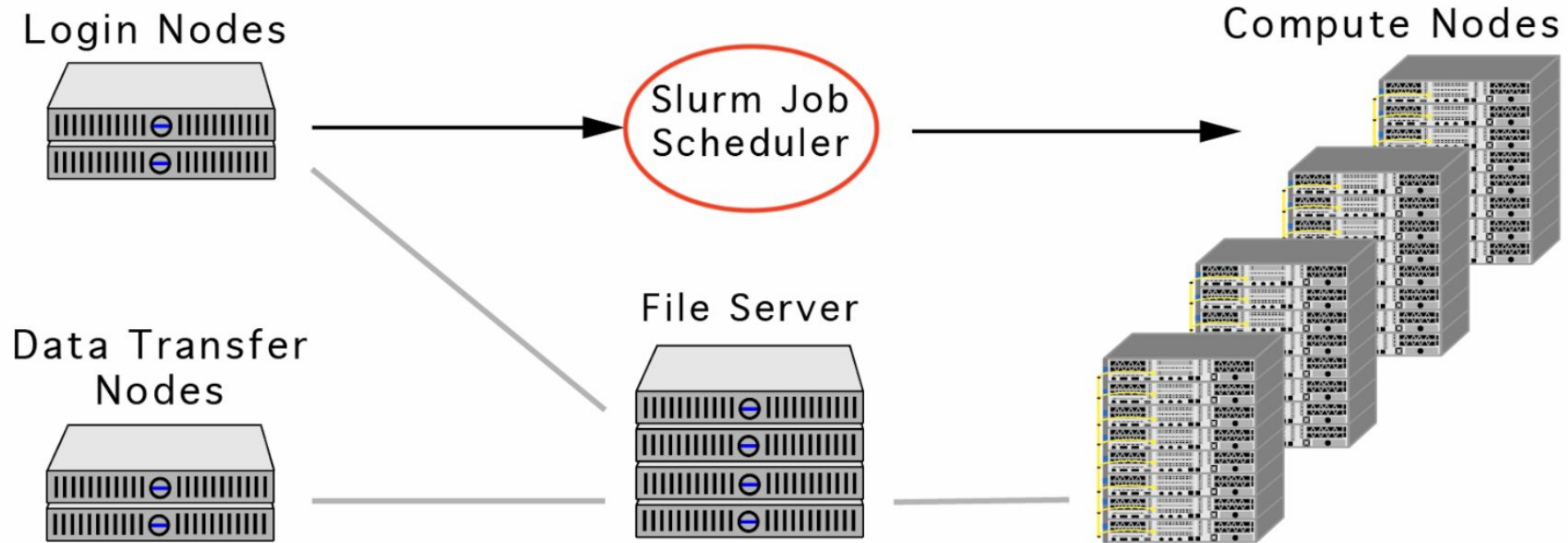
Announcements

- Final Project
 - Piazza - search for teammates
 - Next Friday: Project introduction / handout
- Remember reading quizzes!

Overview

- PA1 Logistics
 - Great Lakes
 - Setup
 - Testing & Grading
- Optimization Tips
 - Pthreads
 - SIMD

Great Lakes



Setup

- [PA1 Handout](#)
- **If you have not requested a Great Lakes login please do so [here](#)**
- Submission Format
 - 1 paragraph report
 - beamform.c

Testing

- Take advantage of submit.sh
 - Provides exclusive node access
 - #SBATCH --cpus-per-task=36
- Local development useful for debugging and quick feedback
- Target 8x speedup

Optimization Strategies- Pthreads

```
/* Now compute reflected distance, find index values, add to image */
for (it_rx = 0; it_rx < trans_x * trans_y; it_rx++) {

    image_pos = image; // Reset image pointer back to beginning
    point = 0; // Reset

    // Iterate over entire image space
    for (it_t = 0; it_t < sls_t; it_t++) {
        for (it_p = 0; it_p < sls_p; it_p++) {
            for (it_r = 0; it_r < pts_r; it_r++) {

                x_comp = rx_x[it_rx] - point_x[point];
                x_comp = x_comp * x_comp;
                y_comp = rx_y[it_rx] - point_y[point];
                y_comp = y_comp * y_comp;
                z_comp = rx_z - point_z[point];
                z_comp = z_comp * z_comp;

                dist = dist_tx[point++] + (float)sqrt(x_comp + y_comp + z_comp);
                index = (int)(dist/idx_const + filter_delay + 0.5);
                *image_pos++ += rx_data[index+offset];
            }
        }
    }

    offset += data_len;
}
```

Optimization Strategies- SIMD

- Utilize vector instructions
- Handle “extra” elements / iterations with scalar code

```
#include <immintrin.h>
__m256 a_vec = _mm256_load_ps(&array[i]);    // Load 8 floats
__m256 b_vec = _mm256_load_ps(&array[i+8]);  // Load next 8 floats
__m256 result = _mm256_sub_ps(a_vec, b_vec); // Subtract 8 pairs
__m256 squared = _mm256_mul_ps(result, result); // Square 8 values
```